STAT/MA 41600
Midterm Exam #1: October 9, 2026
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Midterm Exam #1 Answers

1a. The probability there are exactly 4 winners is $\binom{5{,}000{,}000}{4}(\frac{1}{2{,}000{,}000})^4(\frac{1{,}999{,}999}{2{,}000{,}000})^{4{,}999{,}996}$.

1b. Because $n = 5{,}000{,}000$ is large, $p = \frac{1}{2{,}000{,}000}$ is small, and $np = \frac{5{,}000{,}000}{2{,}000{,}000} = 5/2$ is moderate, we can approximate the number of lottery winners by a Poisson random variable with mean $np = 5/2$. So the approximate probability in **(a)** is $e^{-5/2}(5/2)^4/4! = 0.1336$.
Similarly, the probability there are exactly 4 library visitors is $\binom{50{,}000}{4}(\frac{1}{10{,}000})^4(\frac{9{,}999}{10{,}000})^{49{,}996}$.

The approximate probability of exactly 4 library visitors is $e^{-5}(5)^4/4! = 0.1755$.

2a. The probability their number of rolls agree is $\sum_{x=1}^{\infty}(1/6)(5/6)^{x-1}(1/4)(3/4)^{x-1} = (1/24)\sum_{x=1}^{\infty}(5/8)^{x-1} = (1/24)/(1-5/8) = 1/9$.

2b. The probability that Rhonda takes more rolls than Bernadette is:
$\sum_{y=1}^{\infty}\sum_{x=y+1}^{\infty}(1/6)(5/6)^{x-1}(1/4)(3/4)^{y-1} = \sum_{y=1}^{\infty}(5/6)^y(1/4)(3/4)^{y-1}$
$= (5/6)(1/4)\sum_{y=1}^{\infty}(5/6)^{y-1}(3/4)^{y-1} = (5/24)\sum_{y=1}^{\infty}(5/8)^{y-1} = (5/24)/(1-5/8) = 5/9$.

Alternatively, the probability that one or both of them stop on a round is $(1/6)(3/4) + (1/6)(1/4)+(5/6)(1/4)$; Rhonda continues and Bernadette stops with probability $(5/6)(1/4)$, so the desired probability is $(5/6)(1/4)/((1/6)(3/4) + (1/6)(1/4) + (5/6)(1/4)) = 5/9$.
Similarly, if Rhonda uses an 8-sided die instead of a 6-sided die, and if Bernadette uses a coin instead of a 4-sided die, the probability their number of rolls and flips agree is $\sum_{x=1}^{\infty}(1/8)(7/8)^{x-1}(1/2)(1/2)^{x-1} = (1/16)\sum_{x=1}^{\infty}(7/16)^{x-1} = (1/16)/(1-7/16) = 1/9$.

The probability Rhonda's number of rolls strictly exceed Bernadette's number of flips is:
$\sum_{y=1}^{\infty}\sum_{x=y+1}^{\infty}(1/8)(7/8)^{x-1}(1/2)(1/2)^{y-1} = \sum_{y=1}^{\infty}(7/8)^y(1/2)^y = \sum_{y=1}^{\infty}(7/16)^y = 7/9$.

Alternatively, the probability that one or both of them stop on a round is $(1/8)(1/2) + (1/8)(1/2)+(7/8)(1/2)$; Rhonda continues and Bernadette stops with probability $(7/8)(1/2)$, so the desired probability is $(7/8)(1/2)/((1/8)(1/2) + (1/8)(1/2) + (7/8)(1/2)) = 7/9$.

3. With 21 white socks, 8 black socks, and 4 brown socks, we have:

3a. The probability of 2 socks of each color is $\binom{21}{2}\binom{8}{2}\binom{4}{2}/\binom{33}{6} = 315/9889 = 0.03185$.

3b. The probability that all 6 socks match is $(\binom{21}{6} + \binom{8}{6})/\binom{33}{6} = 1939/39556 = 0.04902$.

3c. The probability of exactly 4 white socks is $\binom{21}{4}\binom{12}{2}/\binom{33}{6} = 2565/7192 = 0.3566$.

Similarly, with 18 white socks, 11 black socks, and 4 brown socks, we have:

3a. The probability of 2 socks of each color is $\binom{18}{2}\binom{11}{2}\binom{4}{2}/\binom{33}{6} = 2295/50344 = 0.04559$.

3b. The probability that all 6 socks match is $(\binom{18}{6} + \binom{11}{6})/\binom{33}{6} = 1359/79112 = 0.01718$.

3c. The probability of exactly 4 white socks is $\binom{18}{4}\binom{15}{2}/\binom{33}{6} = 11475/39556 = 0.2901$.

4. The probability that student is enrolled in a probability course is $(.18)(.15)/((.07)(.42)+(.15)(.25) + (.18)(.15)) = 90/313 = 0.2875$.
Similarly, if a randomly chosen song has fiddle music, the conditional probability that it is a country song is $(.10)(.92)/((.10)(.92)+(.20)(.16)+(.25)(0)+(.45)(.12)) = 46/89 = 0.5169$.

5. If we select 9 bears, then $E(X) = (9)(\frac{20}{30}) = 6$ and $\mathrm{Var}(X) = (9)(\frac{20}{30})(1-\frac{20}{30})(\frac{30-9}{30-1}) = \frac{42}{29} = 1.4483$. **Similarly**, if we select 8 bears, then $E(X) = (8)(\frac{20}{30}) = 16/3$ and $\mathrm{Var}(X) = (8)(\frac{20}{30})(1-\frac{20}{30})(\frac{30-8}{30-1}) = \frac{352}{261} = 1.3487$.

These problems come from: (1) 2015, PS 18; and 2014, PS 18; (2) 2016, PS 16;
(3) Practice Problem Set 20; (4) 2018, PS 5; and 2015, PS 5; (5) 2017, PS 19